

4 (a)	single diode.....M1	[2]
	in series with R OR in series with a.c. supply A1	
(b) (i)1	5.4 V (allow ± 0.1 V)..... A1	
(i)2	$V = iR$ $I = 5.4/1.5 \times 10^3$ C1 $= 3.6 \times 10^{-3}$ A A1	
(i)3	time = 0.027 s A1	[4]
(ii)1	$Q = it$ $= 3.6 \times 10^{-3} \times 0.027$ C1 $= 9.72 \times 10^{-5}$ C A1	
(ii)2	$C = \Delta Q/\Delta V$ (allow C – Q/V for this mark) C1 $= (9.72 \times 10^{-5})/1.2$ $= 8.1 \times 10^{-5}$ F A1	[4]
(c)	line: reasonable shape with less ripple..... B1	[1]
5 (a)	field producing force of 1.0 N m^{-1} on wire OR $B = F/Il \sin \theta$ M1	